

GAYATHRI TIRUMALA

Software Engineer

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PROFESSIONAL SUMMARY

GCP-certified Software Engineer with 4+ years of experience building high-throughput backend APIs, ETL pipelines, and cloud-native data systems across fintech and healthcare domains. Proven ability to reduce latency, improve data reliability, and deliver scalable Python and Java services in Agile environments. Holds Google Cloud Associate Cloud Engineer (ACE) certification with hands-on production experience in BigQuery, Dataflow, and Pub/Sub.

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, JavaScript
- **AI/ML:** Generative AI, LLMs, Prompt Engineering, RAG, AI Agents (LangChain – basic exposure)
- **Data Engineering:** ETL pipelines, Data Processing, Data Warehousing concepts
- **Backend:** REST APIs, FastAPI, Spring Boot, Microservices, CI/CD
- **Frontend:** React.js, JavaScript, HTML, CSS, Responsive Design
- **Cloud:** GCP (BigQuery, Dataflow, Pub/Sub, Cloud Storage, Cloud Functions), AWS (S3, Lambda)
- **Databases:** PostgreSQL, MySQL- query optimization, indexing, schema design
- **Testing:** PyTest, unit testing, integration testing, test-driven development
- **Tools:** Docker, Git, Linux, Jira, Agile/Scrum, code review, Jenkins, Bash, UNIX Commands
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Requests, Pydantic
- **Concepts:** Distributed systems, OOPS, data structures, API design

PROFESSIONAL EXPERIENCE

Software Engineer | Fifth Third Bank | Remote, USA

Jun 2025 – Present

- Engineer and maintain backend APIs for core banking workflows accounts, fund transfers, and transaction processing serving high-volume production traffic.
- Optimize API response times through PostgreSQL query redesign and query plan analysis, meaningfully reducing latency across critical banking endpoints.
- Build automated data pipelines for financial transaction ingestion, reducing manual reconciliation effort and improving downstream reporting accuracy.
- Implemented state management using React Context API for efficient data handling.
- Work on integrating LLM-based features using APIs (OpenAI/Hugging Face) to enhance data processing and user interactions.
- Improve system reliability through structured error handling, input validation, and retry logic, reducing recurring API failures and on-call incidents.
- Maintain high unit test coverage using PyTest; collaborate with frontend and QA teams on API contracts and integration testing.
- Identify and resolve system inefficiencies during sprint retrospectives, contributing to continuous performance improvements across owned services.
- Using Git and CI/CD pipelines for version control and deployment.
- Work closely with backend teams to integrate APIs and ensure smooth data flow.

Software Engineer | Mayo Clinic Healthcare | Remote, USA

Jan 2025 – Jun 2025

- Developed REST APIs for healthcare claims processing and member eligibility workflows, handling large daily record volumes in compliance with HIPAA standards.
- Built ETL pipelines with Python and PostgreSQL to normalize large healthcare datasets, cutting report generation time from 4 hours to under 30 minutes.
- Optimized SQL queries in backend services, achieving significantly faster data retrieval and enabling real-time dashboard updates for clinical operations teams.
- Explored Generative AI use cases and built prototype solutions for internal automation tasks.

- Enforced data quality through automated validation scripts, ensuring accuracy and consistency of records before they reached downstream systems.
- Built responsive user interfaces using React.js and modern JavaScript (ES6+).
- Partnered with clinical, product, and QA teams to deliver system enhancements on schedule within a fast-paced engagement.

Software Engineer | Cybage Software Private Limited | Hyderabad, India **Aug 2021 – Dec 2023**

- Built Python backend services and REST APIs for enterprise clients, supporting ingestion and processing of large, structured datasets across production batch cycles.
- Reduced data retrieval time by 25–30% via SQL index tuning, join restructuring, and N+1 query elimination across three production services.
- Worked on data preprocessing and transformation to support AI/ML workflows.
- Used Bash scripting and UNIX commands for automation and job scheduling.
- Engineered scalable data processing workflows using Pandas and NumPy, replacing manual batch jobs and significantly reducing processing overhead.
- Integrated 5+ third-party APIs (payment, identity, notifications) with retry logic and circuit-breaker patterns for production reliability.
- Contributed to full SDLC across 8 Agile sprints development, unit testing with PyTest, code review, and deployment with consistent on-time delivery.

EDUCATION

Master of Science in Computer Science

University of Central Missouri | **GPA: 3.7** |

Jan 2024 – May 2025

Relevant Coursework: Advanced Algorithms, Advanced Operating Systems, Cloud Computing, Advanced Database Systems, Software Testing & Quality Assurance

Bachelor of Technology in Computer Science & Engineering

IARE Hyderabad | CGPA: 8.44

Aug 2019 – Jun 2023

CERTIFICATION

Google Cloud Associate Cloud Engineer (ACE)